AttendX

Product Requirements Document & Go-To-Market Strategy

From campus portfolio project to a multi-tenant facial-recognition attendance platform

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EXECUTIVE ABSTRACT

AttendX already has the technical shape of a multi-tenant SaaS product — space-based FAISS indexes, join-code onboarding, YuNet detection, FaceNet512 embeddings and anti-spoofing. This report evaluates it as a business: sizing the market, mapping competitors, specifying a PRD, and laying out a phased go-to-market that starts on campus — a segment structurally underserved by enterprise attendance vendors — before expanding to paying commercial users.

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SECTION 1

Executive Summary

AttendX began as a portfolio-grade face-recognition attendance system and has matured into a genuinely multi-tenant SaaS-shaped product: JWT-based auth, per-“space” FAISS indexes in Supabase Storage, join-code onboarding, a fast YuNet detector, FaceNet512 embeddings, and anti-spoofing checks. That is further along than most early-stage attendance tools built by students or first-time founders — the architecture question is mostly answered. The question this report addresses is the one engineering alone cannot answer: **who pays for this, why, and how does it grow from a working system into a defensible business.**

\$3.5B → \$10.2B Global face-recognition attendance market, 2024→2033*	**~16%** CAGR of that category through the early 2030s*	**■20–500** Per-employee/month price band in the Indian market*	**0** Direct campus-native, multi-tenant challengers identified in India

The core thesis

Facial-recognition attendance is not a new category — it is a crowded, commoditizing one dominated by enterprise-workforce vendors (Truein, Matrix COSEC, greytHR, SalaryBox) built for HR/payroll buyers with 50–10,000 employees. None of them are built around the economics and workflows of a **campus**: hundreds of student “spaces” (classes, clubs, hostels, events) that spin up and down constantly, near-zero willingness to pay per seat, and a buyer (a student council, a professor, a club secretary) who is not an HR department. AttendX's join-code, space-based model is accidentally well-suited to this workflow. The recommended path is a **beachhead strategy**: win campus and community-space attendance first, where AttendX is structurally cheaper and faster to deploy than any enterprise competitor, then expand upward into coworking spaces and SME offices once the product has real usage data and testimonials behind it.

What this report recommends

- **Beachhead, not broadside** — target IITH and 2–3 peer campuses first, not the whole facial-recognition-attendance market.
- **Freemium + per-space pricing** — not per-seat SaaS pricing — campus buyers think in terms of “spaces” and events, not headcount contracts.
- **Ship compliance early, not late** — India's DPDP Act treats biometric data as sensitive personal data; consent and deletion flows are a launch requirement, not a v2 feature.
- **Instrument the product before fundraising** — usage data from 2 live campus pilots is worth more, this early, than a polished pitch deck.
- **Treat this as a 12-month portfolio-to-venture bet** — even if AttendX stays a strong resume project rather than a company, this report's structure — PRD, GTM, unit economics — is itself interview-defensible PM material.

SECTION 2

Problem Statement & Market Context

2.1 The problem

Attendance tracking in Indian colleges and community spaces is still overwhelmingly manual: paper registers, verbal roll calls, or, at best, shared spreadsheets and WhatsApp check-ins. This creates three recurring failure modes that a technical buyer (a professor, a placement cell, a club) actually feels: **proxy attendance** (one student marking for another), **administrative drag** (someone manually compiling registers into reports at the end of term), and **no real-time visibility** (no one knows who is actually in the room, which matters for both academic compliance and event/space safety). RFID and fingerprint systems solve some of this in large institutions but are expensive to install per-room and are hygiene- and hardware-dependent. A phone-camera-based, software-only system removes the hardware cost entirely, which is precisely the segment AttendX already targets.

2.2 Market sizing

The broader facial-recognition market is entering a phase of rapid, software-led growth. Independent market research places the global facial-recognition market at roughly USD 10 billion in 2026, growing to over USD 20 billion by 2031 at a compound annual growth rate (CAGR) of around 15–16%, with the software segment (versus cameras/hardware) now taking the majority of that value and the cloud-delivery segment growing fastest of all — both trends favour a lightweight, software-first product like AttendX over hardware-heavy incumbents.

Narrowing to the specific sub-category AttendX competes in, the global face-recognition *attendance system* market specifically was valued at roughly USD 3.5 billion in 2024 and is projected to reach over USD 10 billion by 2033, a CAGR of around 15–17%. Within that, small-and-medium organizations are flagged as the fastest-growing buyer segment, and the education vertical specifically is called out as a growth driver as schools and universities adopt AI-based attendance to cut administrative overhead. In India specifically, the facial-recognition market is smaller in absolute terms (low hundreds of millions of USD today) but is one of the fastest-growing national markets in Asia-Pacific, which is itself the fastest-growing region globally — driven by digitisation programs, smart-campus and smart-city initiatives, and a large, underserved SME and education base.

Metric	Figure	Source (see Sec. 14)
Global facial-recognition market, 2026	~USD 10.0B, growing to ~USD 20.7B by 2031 (CAGR ~15.6%)	MarketsandMarkets
Global face-recognition attendance systems, 2024–2033	USD 3.5B → USD 10.2B (CAGR ~15.8%)	Verified Market Research
India facial-recognition market, 2025–2034	USD 418.5M → USD 903.6M (CAGR ~8.7%)	IMARC Group
Indian attendance-software pricing band	■20–■500 per employee / month	Industry pricing survey, 2025

** Figures are third-party market estimates compiled from publicly available industry research reports (see Section 14 — References). They describe the broader facial-recognition-attendance category; AttendX's addressable slice within campus/community spaces is a small, currently-unquantified subset.*

2.3 Why now

- **Post-pandemic contactless norms stuck** — selfie/face check-in went from a hygiene workaround to a default UX pattern, lowering the trust barrier for new entrants.
- **On-device/edge inference got cheap** — detectors like YuNet and embedding models like FaceNet512 now run fast enough on commodity hardware that a lean team can ship a credible product without an enterprise ML budget — which is exactly AttendX's current stack.
- **India's DPDP Act (2023) is now the operating baseline** — buyers increasingly expect consent flows and audit trails as table stakes, which raises the bar for casual attendance spreadsheets and hobby apps — a real product can differentiate on compliance.
- **Campus tooling budgets remain untapped by incumbents** — enterprise vendors chase corporate HR contracts because per-seat SaaS pricing does not work on a campus; that leaves the segment structurally under-served, not because demand is absent.

SECTION 3

Product Overview — AttendX Today

AttendX v2 is materially more than a face-matching demo. It is worth stating the current architecture plainly, because the GTM strategy in this report is built directly on top of its existing technical shape rather than asking for a rebuild.

3.1 Current architecture snapshot

Layer	Current implementation	Product implication
Tenancy	Multi-tenant with JWT authentication	Already SaaS-shaped — no re-architecture needed to sell to multiple institutions.
Space model	Space-based FAISS indexes in Supabase Storage; join-code onboarding	Maps naturally onto “classes / clubs / events” — the exact unit a campus buyer thinks in.
Detection	YuNet detector (replaced OpenCV Haar/DNN for speed)	Detection is the real latency bottleneck, not matching — correctly prioritised.
Embedding	FaceNet512 with cold-start preloading	Strong accuracy baseline; preloading solves first-request latency, a common demo-killer.
Matching	FAISS (IndexFlat-style per-space index)	Confirmed not the bottleneck at current scale — do not over-invest here yet.

Layer	Current implementation	Product implication
Security	Anti-spoofing / liveness checks	Addresses the single most common objection from any evaluator: “can I hold up a photo?”
Backend	FastAPI + Supabase (Postgres + Storage)	Low infra cost at low scale; will need a scaling plan past a few hundred concurrent spaces.

3.2 What is genuinely differentiated

- **Space-native data model** — most competitors model “employee → company”; AttendX models “face → space,” which is a better fit for anything that is not a fixed employer-employee relationship — classes, events, gyms, co-living, coworking.
- **Zero-hardware onboarding** — join-code based enrolment on any phone camera removes the single biggest adoption barrier for a student club or small institution: capital expenditure.
- **Correctly diagnosed bottleneck** — identifying that detection/inference, not FAISS retrieval, is the latency bottleneck is a non-obvious systems insight that will matter a great deal once the product tries to scale beyond a few hundred concurrent enrolments — it should directly inform the next 2 quarters of engineering investment (see Section 12).

3.3 What is not yet product-market-fit-ready

- No validated paying customer or institution-level deployment outside personal/demo use yet.
- No defined admin experience for a non-technical buyer (a professor or club secretary) to manage a space without engineering hand-holding.
- No explicit DPDP-aligned consent, data-retention, or deletion flow surfaced to end users.
- No usage-based cost model — Supabase/FAISS/compute costs at 50 concurrent institutional spaces are unmodelled.

SECTION 4

Target Users, Personas & Jobs-to-be-Done

A product with a space-based model naturally serves two distinct people per deployment: the **space owner** (who creates and administers a space) and the **member** (who enrols and checks in). GTM sequencing should be read primarily through the space-owner persona, since that is the actual buyer/adopter; members simply need friction-free enrolment.

Persona A — “The Coordinator” — Professor, club secretary, placement-cell coordinator, event organiser (space owner/buyer)

Job to be done	“When I run a recurring class, club, or event, I want an accurate attendance record without manually taking roll call or auditing a spreadsheet, so I can spend that time on the actual session and trust the record if it's ever questioned.”
Current pain	Roll call wastes 5–10 minutes per session; proxy attendance is common and hard to prove; end-of-semester compilation into official records is manual and error-prone.
What AttendX offers	A join-code space that self-enrols members, timestamps every check-in, flags anomalies, and exports a clean report — with zero hardware to procure or install.

Persona B — "The Operator" — Coworking-space manager, gym owner, small-institute admin (adjacent expansion buyer)

Job to be done	"I want to know, in real time, who is actually on my premises without buying and maintaining biometric hardware at every entry point."
Current pain	RFID/fingerprint hardware is capex-heavy and per-door; manual sign-in sheets don't scale past one location; existing SaaS attendance tools price per-seat, which is a bad fit for walk-in / drop-in usage patterns.
What AttendX offers	The same space-based, phone-camera model — already validated on campus — repackaged with usage-based pricing for a paying commercial buyer.

Persona C — "The Member" — Student, employee, gym member, event attendee (end user, not the buyer)

Job to be done	"I just want to check in fast, without carrying a card or repeating a manual process every single time."
Current pain	Queuing at fingerprint scanners; forgetting ID cards; re-entering the same details across multiple unrelated systems.
What AttendX offers	One-time enrolment per space via join code; sub-2-second face check-in on their own device or a shared kiosk.

SECTION 5

Competitive Landscape

The competitive set is real, funded, and has years of enterprise sales motion — but every serious player is built around per-employee, HR-department pricing and workflows. That is a structural gap, not a temporary one, because their entire go-to-market (payroll integrations, compliance reporting for PF/ESI/TDS) is built around a corporate HR buyer that a campus club or coworking space simply does not have.

Player	Core wedge	Pricing signal	Gap AttendX exploits
Truein	Face attendance for contract & distributed workforce (construction, retail, field staff)	~■1–3 / user / month software tier; premium plans ~■390/user/yr + setup fee	Built for HR admins managing headcount, not self-serve space creation by a non-technical coordinator.
Matrix COSEC	Enterprise access-control + attendance + visitor management bundle	Enterprise/hardware-linked, custom quotes	Overkill and over-priced for a single class, club, or small institute; hardware-anchored.
greytHR (Visage)	Selfie + geo-tag attendance bolted onto a 15+ year payroll/compliance engine	Bundled into HRMS suite pricing	Payroll-compliance is the product; attendance is a feature. No concept of ad-hoc "spaces."
SalaryBox	SME-focused all-in-one HRMS with face check-in	~■35–80 / user / month	Targets registered small businesses with a payroll need — not campuses or informal communities.
Manual / spreadsheet / RFID	Status quo across most Indian campuses today	Low direct cost, high hidden labour cost	AttendX's real competitor for the beachhead segment — the bar to beat is "nothing," not a rival SaaS product.

5.1 Positioning map (qualitative)

Where AttendX should sit

On one axis, **deployment weight** (hardware-heavy enterprise devices vs. phone-camera software-only) and on the other, **buyer type** (corporate HR vs. self-serve community admin). Truein, Matrix COSEC and greytHR cluster toward hardware-adjacent, HR-buyer solutions. SalaryBox is the closest existing competitor in “software-only, SME buyer” territory — but still assumes a registered business with payroll needs. AttendX's unclaimed quadrant is **software-only, self-serve community buyer**: a professor, club secretary, or space operator who can create a space and get value in minutes, with no sales call and no per-seat contract.

5.2 Why incumbents are unlikely to respond quickly

- Their sales motion, support structure, and pricing engine are built around enterprise contracts and payroll compliance — retooling for a low-ARPU, high-volume campus segment would cannibalise their own average contract value.
- Campus and community deployments are individually small and fragmented, which is unattractive to a funded competitor optimising for enterprise ACV, but is exactly the kind of segment a lean, low-overhead builder can serve profitably at small scale.

SECTION 6

Product Requirements Document (PRD)

6.1 Product principles

- **Zero-hardware, always** — any requirement that assumes dedicated hardware at the point of check-in should be treated as an enterprise-tier feature, not a default.
- **Space is the atomic unit** — every feature should be scoped to “what does this do inside one space” before it is generalised across an institution.
- **Consent is a first-class object** — not an afterthought — biometric enrolment must always be an explicit, revocable action taken by the member, never the space owner on their behalf.
- **Latency budget protects trust** — check-in must stay under ~2 seconds end-to-end; anything slower reads as “not working” to a non-technical user standing in front of a queue.

6.2 Functional requirements — P0 (must-have for first paying pilot)

ID	Requirement	Rationale
P0-1	Space owner can create a space and generate a join code in under 60 seconds, no onboarding call required	Self-serve is the entire GTM thesis — if it needs a sales call, the unit economics break.
P0-2	Member enrolment requires explicit consent capture (what data, why, retention period) before any face data is stored	DPDP Act compliance floor; also a genuine trust/adoption lever.
P0-3	Member can request deletion of their biometric data and see it actually removed from the space index	Legal requirement and a differentiator vs. competitors who treat this as an edge case.
P0-4	Space owner can export attendance as CSV/PDF filtered by date range	The #1 concrete deliverable a professor or coordinator needs at term-end.

ID	Requirement	Rationale
P0-5	Anti-spoofing check runs on every check-in, not just enrolment	Closes the most common live objection during any pilot demo.
P0-6	Check-in works on a shared kiosk device (single phone/tablet at a room entrance) as well as individual devices	Campuses will want one device per room, not one device per student.
P0-7	Basic anomaly flagging (e.g., same face checking into two spaces in an implausible time window)	Directly targets the proxy-attendance failure mode that is the core pitch.

6.3 Functional requirements — P1 (near-term, unlocks expansion)

ID	Requirement	Rationale
P1-1	Role-based sub-admins within a space (e.g., a TA who can view but not export)	Needed once a single space owner is not the only person managing it.
P1-2	Recurring-schedule spaces (auto-open a check-in window for a recurring class slot)	Removes manual “open/close” actions for the coordinator each session.
P1-3	Institution-level rollup dashboard across many spaces	First real institution-wide selling point once beyond a single club/class pilot.
P1-4	Usage-based billing hooks (space-count or check-in-count metering)	Required before any monetisation beyond a flat pilot fee.
P1-5	Webhook / basic API for export into existing LMS or ERP tools campuses already use	Removes “yet another silo” objection from IT-conscious buyers.

6.4 Non-functional requirements

Category	Target	Notes
Latency	< 2s end-to-end check-in (P95)	Detection (YuNet) + embedding + FAISS match; detection/embedding are the levers, not FAISS.
Accuracy	> 99% true-accept at enrolled-condition lighting; documented false-accept rate	Publish this number once measured — it becomes a trust asset in sales conversations.
Availability	99.5% during institutional operating hours	Lower bar than 24/7 enterprise SLA — appropriate for campus-hours usage patterns.
Data residency & retention	India-region storage; configurable retention with hard-delete on request	DPDP alignment; also a differentiator to state explicitly in marketing.
Cost per active space	Modelled and tracked from pilot #1 onward	Supabase/FAISS/compute cost at scale is currently unmodelled — close this gap before pricing is finalised.

SECTION 7

Use Case Portfolio

Ranked by fit with AttendX's current architecture and by go-to-market difficulty — not by theoretical market size. The goal is sequencing, not breadth.

Use case	Fit today	GTM difficulty	Notes
Classroom / course attendance	High — matches space model exactly	Low (warm network at IITH)	Best pilot candidate: recurring, high-frequency, clear proxy-attendance pain.
Student club / society attendance	High	Low	Fast to pilot; low stakes if something breaks; strong word-of-mouth loop within campus.
Placement Cell / recruiter drives & interviews	High	Low–Medium	Nimish's own FCC/Placement Cell involvement is a built-in distribution channel.
Campus event check-in (fests, workshops)	High	Low	Bursty, one-off spaces — good stress test for FAISS index creation/teardown at volume.
Hostel / mess attendance	Medium	Medium	Higher stakes (linked to fees/allocation); needs stronger accuracy guarantees before pitching.
Coworking space check-in	Medium–High	Medium	First commercial (paying) expansion target after campus proof points.
Gym / fitness studio check-in	Medium	Medium	Similar shape to coworking; smaller ACV but very fast sales cycle.
Small institute / coaching-centre attendance	High	Medium	Adjacent to core campus wedge; parents-as-stakeholders adds a compliance-sensitive angle.
Exam-hall identity verification	Medium	High	High accuracy/liveness bar and institutional risk aversion; treat as v2, not v1.
Corporate/office attendance	Low fit today	High	This is the incumbents' home turf (Truein, greytHR) — avoid until AttendX has clear differentiation beyond price.

> **Explicit non-goal for the next 12 months**
>
> Do not chase corporate office attendance. It is the most crowded, most price-competitive part of the market, dominated by players with payroll-compliance moats AttendX does not have and should not try to build in year one.

SECTION 8

Business Model & Pricing

8.1 Recommended model: freemium, priced per space, not per seat

Because the buyer is a coordinator managing a space rather than an HR department managing headcount, per-seat SaaS pricing (the industry default) is a poor fit — a 300-person lecture and a 6-person club committee should not cost proportionally different amounts to the coordinator who is very likely paying out of pocket or a small club budget. A per-space, tiered model aligns price with the actual unit of adoption.

Tier	Who it's for	Indicative price	What's included
Free	Individual coordinators, clubs, small classes piloting the product	■0	1–3 active spaces, up to ~60 members/space, CSV export, standard check-in.
Campus Pro	Departments, placement cells, recurring-course instructors	Flat institutional or per-department fee (e.g., ■5k–25k/semester, negotiated per pilot)	Unlimited spaces within the institution, rollup dashboard, priority support, anomaly flagging.
Operator	Coworking spaces, gyms, coaching institutes (commercial, paying buyers)	Usage-based: per active space or per check-in volume, benchmarked below Truein/SalaryBox's per-seat rates	API/webhook access, sub-admin roles, SLA-backed support.

Indicative pricing only — to be validated against real willingness-to-pay data from pilot conversations before being finalised; see Section 13.

8.2 Unit economics — what to model before pricing is locked

- Cost per active space per month (Supabase storage + FAISS compute + inference cost at realistic check-in frequency) — currently unmeasured; this is the single most important number to instrument before quoting Operator-tier pricing.
- Marginal cost of a new enrolment (embedding generation + storage) vs. marginal cost of a check-in (detection + inference + match) — these are different cost drivers and should be tracked separately.
- Support cost per institutional pilot — campus pilots are cheap to acquire (warm network) but can be expensive to support if onboarding isn't genuinely self-serve; P0-1 in the PRD exists specifically to keep this cost near zero.

SECTION 9

Go-To-Market Strategy

The GTM sequence below follows a classic beachhead-market playbook: win a small, well-understood, reachable segment completely before expanding — rather than spreading thin across campus, coworking, and enterprise simultaneously.

9.1 Phase 1 (Months 0–3): Prove it inside IIT Hyderabad

- **Target** — 2–3 recurring spaces Nimish already has direct access to — a course, FCC, and/or the Placement Cell — where trust and distribution already exist.
- **Goal** — Not revenue. The goal is a working deployment with real usage data: accuracy in real classroom lighting, actual check-in latency under load, and qualitative feedback from a non-technical coordinator using it without engineering support.
- **Success metric** — ≥ 2 spaces running for a full month with < 5% support-ticket rate and a coordinator willing to give a named testimonial.

9.2 Phase 2 (Months 3–6): Expand within campus, start peer-campus outreach

- **Target** — Additional spaces across IITH (more courses, hostels, events) plus warm introductions to 1–2 peer IITs/NITs through campus network and hackathon/competition contacts.
- **Goal** — Validate that onboarding works without Nimish personally walking someone through it — the real test of the self-serve thesis in P0-1.
- **Success metric** — First institution outside IITH onboarded with zero live-support sessions required during setup.

9.3 Phase 3 (Months 6–12): First paying commercial pilots

- **Target** — 2–3 coworking spaces or coaching institutes in a city with existing network reach (Delhi/NCR is a natural first market given the family's real-estate network at Wadhwa Business Centre, which is itself a plausible first commercial pilot site).
- **Goal** — Convert at least one paying Operator-tier customer and use that engagement to pressure-test the usage-based pricing model in Section 8.
- **Success metric** — ≥ 1 signed paying pilot with a defined renewal decision point.

9.4 Channel strategy

Channel	Fit	Notes
Founder-led warm network (campus, FCC, Placement Cell)	Primary, Phase 1–2	Zero-cost, high-trust; the only channel that matters until there is a testimonial.
Peer-campus / hackathon / competition network	Secondary, Phase 2	Natural expansion path through existing student-community ties.
Inbound content (build-in-public on the technical architecture)	Supporting, all phases	The FAISS-not-the-bottleneck insight and DPDP-compliance angle are genuinely differentiated content, not generic SaaS marketing.
Direct outreach to coworking/coaching operators	Phase 3 only	Should wait until there is at least one campus testimonial to reference — cold outreach without proof points wastes the channel.

SECTION 10

Financial Outlook (Illustrative)

These are directional planning numbers, not forecasts — intended to sanity-check the business model in Section 8, not to be presented as-is to any investor. Every figure should be replaced with real data as soon as Phase 1 pilots produce it.

Milestone	Active spaces	Paying institutions	Illustrative MRR	Primary cost driver
End of Phase 1 (Month 3)	2–5	0 (free pilots)	■0	Founder time; near-zero infra cost at this scale
End of Phase 2 (Month 6)	15–25	0–1	■0–■5k	Supabase/compute cost starts to matter; needs to be measured
End of Phase 3 (Month 12)	40–80	2–4	■15k–■40k	Support cost per institution; infra cost scaling with check-in volume

Illustrative only. Treat Month 12 MRR as a stress-test of the pricing model in Section 8, not a fundraising claim — real numbers depend entirely on Phase 1–2 pilot data.

10.1 What determines whether this is a business or a strong project

- Whether the cost per active space (Section 8.2) stays low enough that Campus Pro pricing (a few thousand rupees per semester) is actually profitable — not just revenue-positive.
- Whether at least one Operator-tier commercial pilot converts and renews, proving willingness to pay outside the warm campus network.
- Whether onboarding genuinely stays self-serve at institution #2 and beyond — if every new customer needs founder hand-holding, this is a services business wearing a SaaS costume.

SECTION 11

Risks, Compliance & Mitigations

Risk	Likelihood	Impact	Mitigation
Biometric data handled without valid consent flow (DPDP Act non-compliance)	Medium	High	Treat P0-2/P0-3 (explicit consent + deletion) as launch-blocking, not post-launch — India's DPDP Act classifies biometric data as sensitive personal data requiring explicit consent.
Spoofing (photo/video/mask) undermines trust in a live pilot demo	Medium	High	Anti-spoofing already implemented (Section 3) — ensure it runs on every check-in, not just enrolment (P0-5), and publish a measured false-accept rate.
Accuracy degrades in real classroom conditions (lighting, masks, angle) vs. controlled testing	Medium	Medium	Measure and publish P95 accuracy from Phase 1 pilots before any institution-wide claims; treat this as a required NFR (Section 6.4), not an assumption.
Onboarding is not actually self-serve, quietly turning this into a services business	Medium	Medium	Track live-support-session count per new space/institution as a core metric (Section 12); P0-1 exists specifically to guard against this.
Incumbent (Truein/SalaryBox) launches a cheap campus-specific tier	Low–Medium	Medium	Their pricing and sales motion are structurally built around per-seat HR contracts (Section 5.2); a fast follow is possible but not their natural next move — monitor, don't over-index on it.
Institutional risk-aversion blocks adoption for higher-stakes use cases (exams, hostel access)	Medium	Low–Medium	Sequencing in Section 7 deliberately defers these use cases until accuracy and trust are proven on lower-stakes spaces first.
Founder bandwidth — this competes directly with academics, placements, and other commitments	High	Medium	Scope Phase 1 tightly (Section 9.1); treat this explicitly as a 12-month side-bet with defined checkpoints, not an open-ended commitment.

11.1 Compliance checklist (DPDP Act alignment)

- Explicit, informed consent captured before any biometric enrolment (what is collected, why, how long it's retained) — not bundled into generic terms of service.
- A visible, working deletion path for any member to remove their biometric data from a space.
- Data residency within India-region infrastructure (already true given the current Supabase region configuration — verify and document this explicitly).
- An audit trail of who accessed or exported attendance data and when, for institutional accountability.

SECTION 12

Metrics, OKRs & Roadmap

12.1 North-star metric

> **North Star**
>
> **Weekly active spaces with ≥ 1 real check-in** — not total signups, not total faces enrolled. This is the single metric that reflects genuine recurring usage rather than one-time curiosity, and it is the number to report to yourself every week during Phase 1–2.

12.2 Supporting metrics by phase

Phase	Primary metric	Target	Guardrail metric
Phase 1 (0–3mo)	Weekly active spaces	≥ 2 sustained for 4 consecutive weeks	Support tickets per space < 1/week
Phase 2 (3–6mo)	Spaces onboarded without live founder support	≥ 50% of new spaces	Check-in P95 latency < 2s at higher concurrency
Phase 3 (6–12mo)	Paying institutions retained past first billing cycle	≥ 1	Cost per active space tracked and < price charged

12.3 12-month roadmap

Quarter	Engineering focus	GTM focus
Q1 (M1–3)	Ship P0 requirements (Section 6.2): consent flow, deletion, kiosk mode, anomaly flagging, export	Launch Phase 1 campus pilots (Section 9.1)
Q2 (M4–6)	Harden self-serve onboarding; begin instrumenting cost-per-space (Section 8.2); start P1 recurring-schedule spaces	Expand within IITH; first peer-campus outreach (Section 9.2)
Q3 (M7–9)	Institution-level rollup dashboard (P1-3); usage-based billing hooks (P1-4)	First commercial Operator-tier conversations (Section 9.3)
Q4 (M10–12)	API/webhook access (P1-5); scale-test detection/inference pipeline past current concurrency ceiling	Close ≥ 1 paying pilot; reassess venture-vs-portfolio-project decision (Section 13)

SECTION 13

Recommendation & Next 90 Days

AttendX does not need a new architecture to become a real product — it needs a narrower, sharper go-to-market than “face recognition attendance system,” and a small set of P0 requirements that turn a working demo into something a non-technical coordinator can adopt without help. The recommendation is to treat the next 90 days as a deliberate, bounded experiment rather than either a full startup commitment or an abandoned side-project.

Next 90 days, concretely

- **Weeks 1–2** — Ship the consent-capture and deletion flow (P0-2, P0-3). This is both the highest-leverage compliance work and a genuine trust differentiator to lead with in any pilot conversation.
- **Weeks 2–4** — Identify and confirm 2–3 Phase-1 pilot spaces inside IITH (a course, FCC, or Placement Cell activity) and ship kiosk mode (P0-6) if any of them need shared-device check-in.
- **Weeks 4–12** — Run the pilots live; instrument weekly active spaces (Section 12.1) and support-ticket rate as the two numbers that matter; measure real accuracy and latency under classroom conditions, not synthetic tests.
- **Week 12 checkpoint** — Review against the Phase 1 success metric in Section 9.1. If met, move to Phase 2 outreach; if not, diagnose whether the gap is technical (accuracy/latency), product (onboarding friction), or distribution (wrong initial spaces chosen) before changing anything else.

How to think about this either way

If AttendX becomes a genuine part-time venture, this report's PRD and GTM sequencing is the actual operating plan. If it stays a portfolio project, the process behind this report — market sizing, competitive positioning, persona-driven requirements, phased GTM, unit economics, and a risk register — is exactly the artefact that demonstrates product thinking in an ML engineering or PM interview, independent of whether the business itself ever launches. Both outcomes justify the next 90 days of work.

SECTION 14

References

Global facial-recognition market sizing — MarketsandMarkets, “Facial Recognition Market Report 2026–2031.”

Global face-recognition attendance system market sizing — Verified Market Research, “Face Recognition Attendance System Market Size, Growth, Trends & Forecast 2033.”

India facial-recognition market sizing — IMARC Group, “India Facial Recognition Market Size & Forecast 2034.”

Global facial-recognition market sizing (cross-check) — Fortune Business Insights and Coherent Market Insights facial-recognition market reports, 2026 editions.

Competitor positioning and pricing signals — Truein, Matrix COSEC, greytHR, and SalaryBox public pricing and product pages; Techjockey and SelectHub vendor comparisons, 2025–2026.

Indian attendance-software pricing benchmarks — industry buyer-guide research compiled 2025–2026.

DPDP Act (Digital Personal Data Protection Act, 2023) biometric-data classification — as summarised in current industry compliance guidance for attendance-software vendors, 2026.

All figures are third-party estimates current as of research conducted July 2026 and should be re-verified before use in any external-facing fundraising or partnership document.